Guided LAB 331.2.7 - Basic Logical functions in Excel

Lab Overview:

In this lab, learners will explore and practice using Excel's basic logical functions, focusing on the **`IF()`**, **`SUMIF()`**, **`AND()`**, and **`OR()`** functions. These functions allow users to create dynamic conditions in spreadsheets, essential for data analysis and decision-making processes. By working through real-world examples, such as categorizing Pokémon types, evaluating scores, and calculating commissions, learners will build the skills needed to effectively apply these functions in various scenarios. The exercises cover both single-condition and multi-condition logical functions, guiding learners through both simple and nested uses of each function.

Learning Objective:

By the end of this lab, learners will be able to:

- Apply Excel's logical functions, including `IF()`, `SUMIF()`, `AND()`, and `OR()`, to perform conditional data analysis.
- Describe the purpose and syntax of basic and nested Excel logical functions.
- Analyze complex datasets to determine the appropriate function or combination of functions for accurate data insights.
- **Create** nested logical functions to handle multiple conditions within a single formula, and describe how to combine `IF()` with `AND()` or `OR()` for complex decision logic.

DataSet Overview:

In this lab, we will utilize the Poke Mart Shop Cart dataset for the practice.

➔ **Click here to download the Excel file for the dataset.** The excel file has multipleSample_LAB 331.2.7.zip sheets.

1: Example: IF() Function

The **IF()** function is a predefined function in Excel, which returns values based on a true or false condition.

Syntax: `=IF()` has 3 parts:

```
=IF(logical_test, [value_if_true], [value_if_false])
```

The condition is referred to as `logical_test`, which can check things like:

- If a number is greater than another number `>`.
- If a number is smaller than another number `<`.
- If a number or text is equal to something `=`.

Note: You can define both the **condition** and the **return values** when using the function.

Also, be aware that different parts of the function are separated by symbols such as a **comma (,)** or a **semicolon (;)**, depending on your **regional language settings**:

- If your Excel is set to **English (United States)**, use **commas** to separate arguments.
  Example: `=IF(A1>10, "Yes", "No")`

- If your Excel is set to a **language like German**, use **semicolons** instead.
  Example: `=IF(A1>10; "Yes"; "No")`

1.1: Example IF() function (equal to)

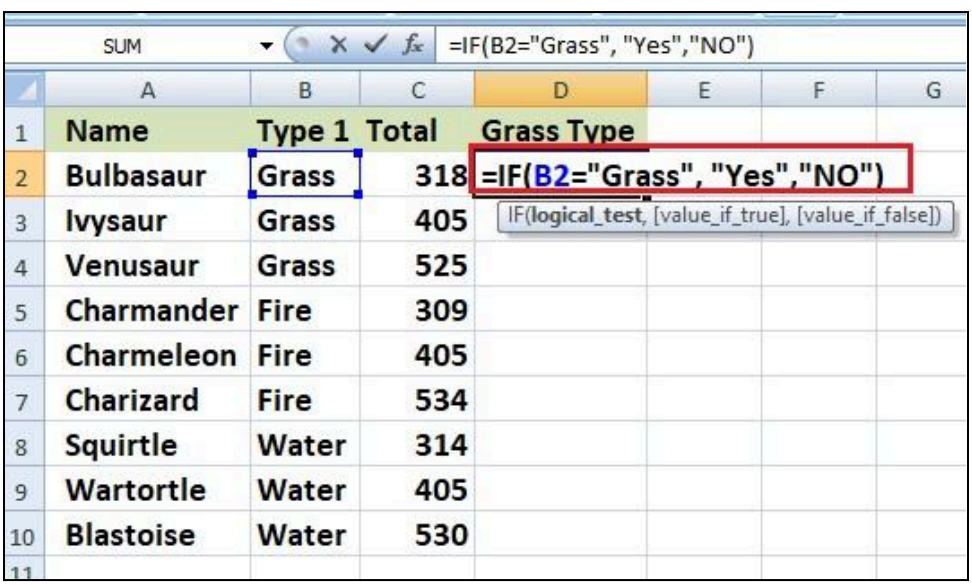
In this example, we will check if the `Pokemon` type is grass:

- The condition is if the **"Type 1"** value for the Pokémon is **"Grass."**
- The function returns "Yes" or "No."

	A	B	C	D	E
1	Name	Type 1	Total	Grass Type	
2	Bulbasaur	Grass	318		
3	Ivysaur	Grass	405		
4	Venusaur	Grass	525		
5	Charmander	Fire	309		
6	Charmeleon	Fire	405		
7	Charizard	Fire	534		
8	Squirtle	Water	314		
9	Wartortle	Water	405		
10	Blastoise	Water	530		

Step 1. Select the cell `D2`.
Step 2. Type `=IF`
Step 3. Double-click the IF command.
Step 4. Specify the condition `B2="Grass"`
Step 5. Type `,`
Step 6. Specify the value `"Yes"` for when the condition is TRUE.
Step 7. Type `,`
Step 8. Specify the value `"No"` for when the condition is FALSE.
Step 9. Hit enter.

The final function will be **=IF(B2="Grass", "Yes","NO")**



Since the value in cell B2 is "Grass," the condition is **true**, and the function will return "**Yes**."

Note: Text values needs to be in quotes: " "

The function can be repeated with the **filling function** for each row to perform the same check for each Pokémon, as shown in the screenshot below.

	A	B	C	D	E
1	Name	Type 1	Total	Grass Type	
2	Bulbasaur	Grass	318	=IF(B2="Grass"; "Yes"; "No")	
3	Ivysaur	Grass	405	=IF(B3="Grass"; "Yes"; "No")	
4	Venusaur	Grass	525	=IF(B4="Grass"; "Yes"; "No")	
5	Charmander	Fire	309	=IF(B5="Grass"; "Yes"; "No")	
6	Charmeleon	Fire	405	=IF(B6="Grass"; "Yes"; "No")	
7	Charizard	Fire	534	=IF(B7="Grass"; "Yes"; "No")	
8	Squirtle	Water	314	=IF(B8="Grass"; "Yes"; "No")	
9	Wartortle	Water	405	=IF(B9="Grass"; "Yes"; "No")	
10	Blastoise	Water	530	=IF(B10="Grass"; "Yes"; "No")	
11				IF (logical_test; [value_if_true]; [value_if_false])	
12					

Now, each row has a check for Grass Type.

	A	B	C	D	E
1	Name	Type 1	Total	Grass Type	
2	Bulbasaur	Grass	318	Yes	
3	Ivysaur	Grass	405	Yes	
4	Venusaur	Grass	525	Yes	
5	Charmander	Fire	309	No	
6	Charmeleon	Fire	405	No	
7	Charizard	Fire	534	No	
8	Squirtle	Water	314	No	
9	Wartortle	Water	405	No	
10	Blastoise	Water	530	No	
11					

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	A	B	C	D	E
1	Name	Type 1	Total	Grass Type	
2	Bulbasaur	Grass	318	Yes	
3	Ivysaur	Grass	405	Yes	
4	Venusaur	Grass	525	Yes	
5	Charmander	Fire	309	No	
6	Charmeleon	Fire	405	No	
7	Charizard	Fire	534	No	
8	Squirtle	Water	314	No	
9	Wartortle	Water	405	No	
10	Blastoise	Water	530	No	
11					
12					
13	Step 1. `=IF(B2="Grass","Yes","No")				
14	*if this logical test is true the function returns customized responses for TRUE and FALSE				
15	Step 2. Drag down to autofill				

1.2: Example IF() Function (greater than)

Let's check if the Pokémon's total **stats are greater than 500**:

- The **if()** function returns "Yes" or "No"

	A	B	C	D	E
1	Name	Type 1	Total	More than 500 Total Stats	
2	Bulbasaur	Grass	318		
3	Ivysaur	Grass	405		
4	Venusaur	Grass	525		
5	Charmander	Fire	309		
6	Charmeleon	Fire	405		
7	Charizard	Fire	534		
8	Squirtle	Water	314		
9	Wartortle	Water	405		
10	Blastoise	Water	530		
11					

Example **IF()** function, step by step:

1. Select the cell `D2`
2. Type `=IF`

3. Double click the IF command.
4. Specify the condition `C2>500`
5. Type `,`
6. Specify the value `"Yes"` for when the condition is TRUE.
7. Type `,`
8. Specify the value `"No"` for when the condition is FALSE.
9. Hit enter.

The final function will be **`=IF(C2>500,"YES","NO")`**.

	A	B	C	D	E
1	Name	Type 1	Total	More than 500 Total Stats	
2	Bulbasaur	Grass	318	=IF(C2>500; "Yes"; "No"	
3	Ivysaur	Grass	405	IF (logical_test; [value_if_true]; [value_if_false])	
4	Venusaur	Grass	525		
5	Charmander	Fire	309		
6	Charmeleon	Fire	405		
7	Charizard	Fire	534		
8	Squirtle	Water	314		
9	Wartortle	Water	405		
10	Blastoise	Water	530		
11					

Since the value in cell `C2` is 318, the condition is false, and the function will return "No."

The function can be repeated with the filling function for each row to perform the same check for each Pokémon, as shown in the screenshot below.

	A	B	C	D	E
1	Name	Type 1	Total	More than 500 Total Stats	
2	Bulbasaur	Grass	318	=IF(C2>500; "Yes"; "No")	
3	Ivysaur	Grass	405	=IF(C3>500; "Yes"; "No")	
4	Venusaur	Grass	525	=IF(C4>500; "Yes"; "No")	
5	Charmander	Fire	309	=IF(C5>500; "Yes"; "No")	
6	Charmeleon	Fire	405	=IF(C6>500; "Yes"; "No")	
7	Charizard	Fire	534	=IF(C7>500; "Yes"; "No")	
8	Squirtle	Water	314	=IF(C8>500; "Yes"; "No")	
9	Wartortle	Water	405	=IF(C9>500; "Yes"; "No")	
10	Blastoise	Water	530	=IF(C10>500; "Yes"; "No")	
11				IF (logical_test; [value_if_true]; [value_if_false])	
12					

Now, each row is checked for having more than 500 total stats.

	A	B	C	D
1	Name	Type 1	Total	More than 500 Total Stats
2	Bulbasaur	Grass	318	No
3	Ivysaur	Grass	405	No
4	Venusaur	Grass	525	Yes
5	Charmander	Fire	309	No
6	Charmeleon	Fire	405	No
7	Charizard	Fire	534	Yes
8	Squirtle	Water	314	No
9	Wartortle	Water	405	No
10	Blastoise	Water	530	Yes

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	A	B	C	D
21	Name	Type 1	Total	More than 500 Total Stats
22	Bulbasaur	Grass	318	No
23	Ivysaur	Grass	405	No
24	Venusaur	Grass	525	Yes
25	Charmander	Fire	309	No
26	Charmeleon	Fire	405	No
27	Charizard	Fire	534	Yes
28	Squirtle	Water	314	No
29	Wartortle	Water	405	No
30	Blastoise	Water	530	Yes
31				
32				
33	Step 1. `=IF(C22>500,"Yes","No")			
34	*can use math operators in the logical test			

1.3: Example: Nested IF() to Check a Sequence of Conditions

The Excel **IF()** Function can take up to 64 conditions at once. While it's inadvisable to create long Nested IF functions, in the case of limited conditions, you can create a formula that checks conditions in a sequence.

In the example below, we check for two conditions.

- The first condition checks whether the marks are less than 35. If this is **TRUE,** it returns **Fail**.

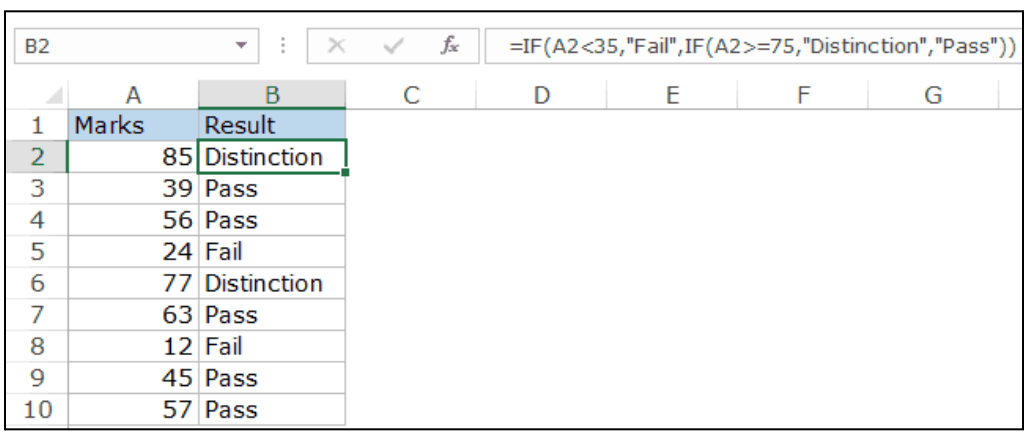
- In case the first condition is **FALSE**, which means that the student scored above or equal to **35**, it checks for another condition. It checks if the **marks are greater than or equal to 75**. If this is **true**, it returns **Distinction;** else, it simply returns **Pass**.

Condition	**Result**
Marks < 35	Fail
Marks >= 35 and Marks < 75	Pass
Marks >= 75	Distinction

In this example we will use the sheet name `***1.3_NestedIF***` from the sample file.

The final function is mentioned below:

```
=IF(A2<35, "Fail", IF(A2>=75, "Distinction", "Pass"))
```



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	A	B	C	D	E
1	Mark	Result			
2	85	Distinction			
3	39	Pass			
4	56	Pass			
5	24	Fail			
6	77	Distinction			
7	63	Pass			
8	12	Fail			
9	45	Pass			
10	57	Pass			
11					
12					
13	Step 1. `=IF(A2<35, "Fail", IF(A2>=75,"Distinction","Pass"))				
14	*Nested IFS test for multiple conditions				
15	*Need to close the parentheses for both open statements				

1.4: Example: Calculating Commissions Using Nested IF() Function

The Excel If() Function allows you to perform calculations in the value section. A good example of this is calculating the sales commission for a sales rep using the IF function.

In the example below, a sales rep gets no commission if the sales are less than 50K, gets a 2% commission if the sales are between 50-100K and gets a 4% commission if the sales are more than 100K.

In this example we will use the sheet name `***1.4_NestedIF***` from the sample file.

Below is the final function: The task can be accomplished with the following **OR()** function:

```
=IF(B2<50,0,If(B2<100,B2*2%,B2*4%))
```

C2 =IF(B2<50,0,IF(B2<=100,B2*2%,B2*4%))

	A	B	C	D	E	F	G
1	**Sales Rep**	**Sales**	**Commission**				
2	Mark	89	1.78				
3	Jane	138	5.52				
4	Bob	162	6.48				
5	Tom	134	5.36				
6	Bill	43	0				
7	James	175	7				
8	Harry	37	0				
9	Kevin	45	0				
10	Robert	126	5.04				
11	Martha	55	1.1				

Note: When the sales value is between 50-100K, it returns B2*2%, which is the 2% commission based on the sales value.

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	A	B	C	D	E	F	G
1	Sales Rep	Sales	Commision				
2	Mark	89	1.78				
3	Jane	138	5.52				
4	Bob	162	6.48				
5	Tom	134	5.36				
6	Bill	43	0				
7	James	175	7				
8	Harry	37	0				
9	Kevin	45	0				
10	Robert	126	5.04				
11	Martha	55	1.1				
12		1000					
13							
14							
15	Step 1. `=IF(B2<50,0,IF(B2<100,B2*2%,B2*4%))						
16	*This function is calculating the commision the sales reps earned/did not earn						
17	*the formula looks complex at first glance but learning the syntax is helping with nested IFS						
18	Step 2. Dragged down to Autofill						
19							

2: Example SUMIF() function

The **SUMIF()** function is a predefined function in Excel, which calculates the sum of values in a range based on a true or false condition.

The Excel **SUMIF()** function returns the sum of cells that meet a single condition. Criteria can be applied to **dates, numbers,** and **text**. The **SUMIF()** function supports logical operators **(>,<,<>,=)** and wildcards **(*,?)** for partial matching.

Syntax:

`=SUMIF(range,criteria,[sum_range])`

- `Range` - Range to apply criteria to.

- **`[sum_range]`** - is the range where the function calculates the sum. If it can be optional, cells in range are summed.
- **`criteria`**, which can check things like:
  - If a number is greater than another number `>`
  - If a number is smaller than another number `<`
  - If a number or text is equal to something `=`

Let's find the sum of Total stats for **"Grass"** type Pokemon. The condition is that the type is **"Grass."** In this example we will use the sheet name ***`2_SUMIF`*** from the sample file.

	A	B	C	D	E	F	G
1	Name	Type 1	Total				
2	Bulbasaur	Grass	318		Type	Total Sum	
3	Ivysaur	Grass	405		Grass		
4	Venusaur	Grass	525		Fire		
5	Charmander	Fire	309		Water		
6	Charmeleon	Fire	405				
7	Charizard	Fire	534				
8	Squirtle	Water	314				
9	Wartortle	Water	405				
10	Blastoise	Water	530				

Both below functions will give you the same result:

```
=SUMIF(B2:B10,E3,C2:C10)
```

```
=SUMIF(B2:B11,"Grass",C2:C11)
```

	A	B	C	D	E	F	G	H	I
1	Name	Type 1	Total						
2	Bulbasaur	Grass	318		Type	Total Sum			
3	Ivysaur	Grass	405		Grass	=SUMIF(B2:B10; E3; C2:C10			
4	Venusaur	Grass	525		Fire	SUMIF (range; criteria; [sum_range])			
5	Charmander	Fire	309		Water				
6	Charmeleon	Fire	405						
7	Charizard	Fire	534						
8	Squirtle	Water	314						
9	Wartortle	Water	405						
10	Blastoise	Water	530						
11									

- The function now sums the Total stats for Grass type Pokemon.

- The function can be repeated for the other Pokemon types to compare them, as shown in the screenshot below.

	A	B	C	D	E	F	G
1	Name	Type 1	Total				
2	Bulbasaur	Grass	318		Type	Total Sum	
3	Ivysaur	Grass	405		Grass	=SUMIF(B2:B10; E3; C2:C10)	
4	Venusaur	Grass	525		Fire	=SUMIF(B2:B10; E4; C2:C10)	
5	Charmander	Fire	309		Water	=SUMIF(B2:B10; E5; C2:C10)	
6	Charmeleon	Fire	405			SUMIF (range; criteria; [sum_range])	
7	Charizard	Fire	534				
8	Squirtle	Water	314				
9	Wartortle	Water	405				
10	Blastoise	Water	530				
11							

Note: You can use the **filling function** for the other rows, but make sure to use absolute references for the ranges.

Now, we can see the sum of total stats for the different types of Pokemon.

	A	B	C	D	E	F	G
1	Name	Type 1	Total				
2	Bulbasaur	Grass	318		Type	Total Sum	
3	Ivysaur	Grass	405		Grass	1248	
4	Venusaur	Grass	525		Fire	1248	
5	Charmander	Fire	309		Water	1249	
6	Charmeleon	Fire	405				
7	Charizard	Fire	534				
8	Squirtle	Water	314				
9	Wartortle	Water	405				
10	Blastoise	Water	530				
11							

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	A	B	C	D	E	F
1	Name	Type 1	Total	Grass Type		
2	Bulbasaur	Grass	318	Yes	Type	Total SUM
3	Ivysaur	Grass	405	Yes	Grass	1248
4	Venusaur	Grass	525	Yes	Fire	1248
5	Charmander	Fire	309	No	Water	1249
6	Charmeleon	Fire	405	No		
7	Charizard	Fire	534	No		
8	Squirtle	Water	314	No		
9	Wartortle	Water	405	No		
10	Blastoise	Water	530	No		
11						
12	Step 1. `=SUMIF(B2:C10,E3,C2:C10)					
13	*I chose to use the formula using a cell reference rather than typing "Grass" to avoid having to rewrite the formula to generate the fields for 'Fire' and 'Water'					
14	Step 2. Drag to Autofill for 'Fire' and 'Water'					
15						
16						

3: Example AND() Function:

The **AND()** function is a predefined function in Excel, which returns **TRUE** or **FALSE** based on two or more conditions.

Note: The **AND()** function is often used together with the **IF()** function.

Syntax:

```
=AND([logical1], [logical2], ...)
```

- **logical1—**the first condition that you want to evaluate for TRUE or FALSE.
- **[logical2]** – (Optional) This is the second condition that you want to evaluate for TRUE or FALSE.

3.1: Example AND() Function without IF()

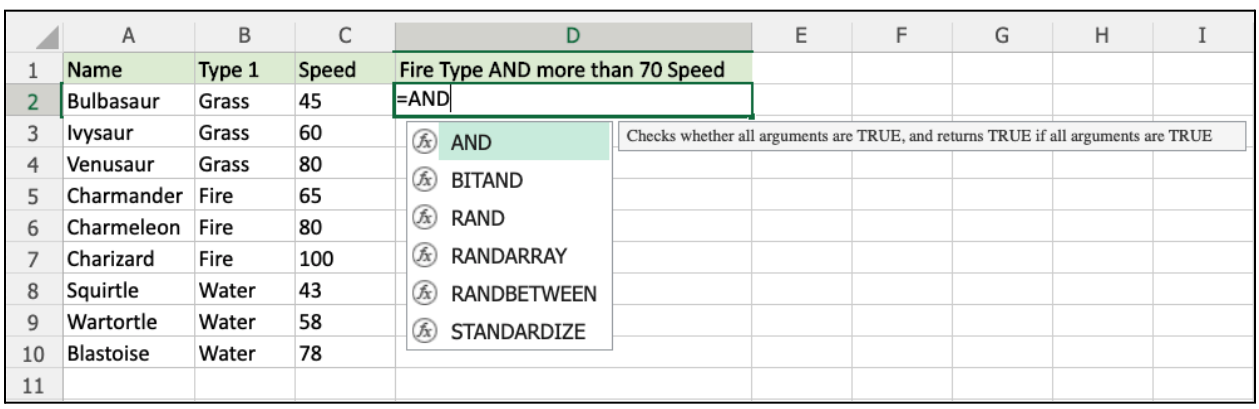
In this example we will use the `**3.1_AND**` sheet from our Sample file.

	A	B	C	D	E
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed	
2	Bulbasaur	Grass	45		
3	Ivysaur	Grass	60		
4	Venusaur	Grass	80		
5	Charmander	Fire	65		
6	Charmeleon	Fire	80		
7	Charizard	Fire	100		
8	Squirtle	Water	43		
9	Wartortle	Water	58		
10	Blastoise	Water	78		

Let's check if the Pokemon type is **fire** and **has speed greater than 70**: The function returns **"TRUE"** or **"FALSE."**

Here is a step-by-step guide:

1. Select the cell `D2`
2. Type `=AND`
3. Double click the AND command.



4. Specify the first condition `B2="Fire"`
5. Type `,`
6. Specify the second condition `C2>70`
7. Hit enter.

Below is the final function: The task can be accomplished with the following **AND()** function:

```
=AND(B2="Fire",C2>70)
```

	A	B	C	D	E
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed	
2	Bulbasaur	Grass	45	=AND(B2="Fire"; C2>70	
3	Ivysaur	Grass	60	AND (logical1; [logical2]; [logical3]; ...)	
4	Venusaur	Grass	80		
5	Charmander	Fire	65		
6	Charmeleon	Fire	80		
7	Charizard	Fire	100		
8	Squirtle	Water	43		
9	Wartortle	Water	58		
10	Blastoise	Water	78		
11					

- Since the value in cell B2 is not "Fire," the first condition is FALSE.
- Since the value in cell C2 is less than 70, the second condition is also FALSE.
- All conditions need to be TRUE for the AND function to return TRUE.

The function can be repeated with the filling function for each row to perform the same check for each Pokémon, as shown in the screenshot below.

	A	B	C	D	E
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed	
2	Bulbasaur	Grass	45	=AND(B2="Fire"; C2>70)	
3	Ivysaur	Grass	60	=AND(B3="Fire"; C3>70)	
4	Venusaur	Grass	80	=AND(B4="Fire"; C4>70)	
5	Charmander	Fire	65	=AND(B5="Fire"; C5>70)	
6	Charmeleon	Fire	80	=AND(B6="Fire"; C6>70)	
7	Charizard	Fire	100	=AND(B7="Fire"; C7>70)	
8	Squirtle	Water	43	=AND(B8="Fire"; C8>70)	
9	Wartortle	Water	58	=AND(B9="Fire"; C9>70)	
10	Blastoise	Water	78	=AND(B10="Fire"; C10>70)	
11					

Now, each row is checked for **Fire** Type and Speed greater than 70:

	A	B	C	D	E
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed	
2	Bulbasaur	Grass	45	FALSE	
3	Ivysaur	Grass	60	FALSE	
4	Venusaur	Grass	80	FALSE	
5	Charmander	Fire	65	FALSE	
6	Charmeleon	Fire	80	TRUE	
7	Charizard	Fire	100	TRUE	
8	Squirtle	Water	43	FALSE	
9	Wartortle	Water	58	FALSE	
10	Blastoise	Water	78	FALSE	
11					

Only **Charmeleon** and **Charizard** both have a `**Fire**` type and **speed greater than 70,** so the function returns **"TRUE."**

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	A	B	C	D	E	F	G	H
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed				
2	Bulbasaur	Grass	45	FALSE				
3	Ivysaur	Grass	60	FALSE				
4	Venusaur	Grass	80	FALSE				
5	Charmander	Fire	65	FALSE	* highlighted the cells that returned TRUE			
6	Charmeleon	Fire	80	TRUE				
7	Charizard	Fire	100	TRUE				
8	Squirtle	Water	43	FALSE				
9	Wartortle	Water	58	FALSE				
10	Blastoise	Water	78	FALSE				
11								
12	Step 1. `=AND(B2="Fire",C2>70)							
13	*if both conditions are TRUE the function returns TRUE							
14	*if one or none of the conditions are FALSE the function returns FALSE							

3.2: Example AND Function with IF()

In Excel, the **AND()** function itself returns either **TRUE** or **FALSE** and these return values CANNOT be changed. However, when you use **AND()** within an **IF()** function, you can customize the output based on the result of the AND() function. For example, you can return “YES” or “NO,” “Open” or “Close,” or any other values you choose.

In this example we will use the **3.2_AND** sheet from our sample file.

Below is the final function: The task can be accomplished with the following **AND()** function:

```
=IF(AND(B2="Fire",C2>70),"YES","NO")
```

	A	B	C	D	E
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed	
2	Bulbasaur	Grass	45	=IF(AND(B2="Fire"; C2>70); "Yes"; "No"	
3	Ivysaur	Grass	60	IF (logical_test; [value_if_true]; [value_if_false])	
4	Venusaur	Grass	80		
5	Charmander	Fire	65		
6	Charmeleon	Fire	80		
7	Charizard	Fire	100		
8	Squirtle	Water	43		
9	Wartortle	Water	58		
10	Blastoise	Water	78		
11					

The function can be repeated with the filling function for each row to perform the same check for each Pokémon, as shown in the screenshot below.

	A	B	C	D	E
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed	
2	Bulbasaur	Grass	45	=IF(AND(B2="Fire"; C2>70); "Yes"; "No")	
3	Ivysaur	Grass	60	=IF(AND(B3="Fire"; C3>70); "Yes"; "No")	
4	Venusaur	Grass	80	=IF(AND(B4="Fire"; C4>70); "Yes"; "No")	
5	Charmander	Fire	65	=IF(AND(B5="Fire"; C5>70); "Yes"; "No")	
6	Charmeleon	Fire	80	=IF(AND(B6="Fire"; C6>70); "Yes"; "No")	
7	Charizard	Fire	100	=IF(AND(B7="Fire"; C7>70); "Yes"; "No")	
8	Squirtle	Water	43	=IF(AND(B8="Fire"; C8>70); "Yes"; "No")	
9	Wartortle	Water	58	=IF(AND(B9="Fire"; C9>70); "Yes"; "No")	
10	Blastoise	Water	78	=IF(AND(B10="Fire"; C10>70); "Yes"; "No")	
11					

Now each row is checked for both being `**Fire**` type and **Speed greater than 70:**

	A	B	C	D	E
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed	
2	Bulbasaur	Grass	45	No	
3	Ivysaur	Grass	60	No	
4	Venusaur	Grass	80	No	
5	Charmander	Fire	65	No	
6	Charmeleon	Fire	80	Yes	
7	Charizard	Fire	100	Yes	
8	Squirtle	Water	43	No	
9	Wartortle	Water	58	No	
10	Blastoise	Water	78	No	
11					

Only **Charmeleon** and **Charizard** both have Fire type and speed greater than 70, so the function returns "Yes."

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	A	B	C	D	E	F
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed		
2	Bulbasaur	Grass	45	No		
3	Ivysaur	Grass	60	No		
4	Venusaur	Grass	80	No		
5	Charmander	Fire	65	No		
6	Charmeleon	Fire	80	Yes		
7	Charizard	Fire	100	Yes		
8	Squirtle	Water	43	No		
9	Wartortle	Water	58	No		
10	Blastoise	Water	78	No		
11						
12	Step 1. `=IF(AND(B2="Fire",C2>70), "Yes","No")					
13	*This was a little tricky to understand at first					
14	*The AND function is nested into the IF function serving as the logical function					
15	*Nesting the AND into the IF function allows customization of the response rather than just TRUE/FALSE					
16	Step 2. Drag down to AutoFill					
17	Step 3. Make sure the formula is accurate all the way down					

4: Example: OR() Function

The **OR()** function is a predefined function in Excel, which is a logical function. The **OR()** function tests if one or more conditions are TRUE and FALSE if all conditions are false. It can evaluate up to 255 expressions.

Syntax:

`OR(logical1, [logical2], ...)`

- `logical1`: The first condition to test.
- `[logical2]`: Additional conditions to test (up to 255 conditions).
- The conditions are referred to as `[logical1], [logical2]`, and so on.

The conditions can check things like:

- If a number is greater than another number `>`
- If a number is smaller than another number `<`
- If a number or text is equal to something `=`

4.1: Example OR() Function without IF()

Let's check if the **Pokémon** type is **water** OR **has defense greater than 60**:

In this example we will use the `**4.1_AND**` sheet from our Sample file.

	A	B	C	D	E
1	Name	Type 1	Defense	Water Type OR more than 60 Defense	
2	Bulbasaur	Grass	49		
3	Ivysaur	Grass	63		
4	Venusaur	Grass	83		
5	Charmander	Fire	43		
6	Charmeleon	Fire	58		
7	Charizard	Fire	78		
8	Squirtle	Water	65		
9	Wartortle	Water	80		
10	Blastoise	Water	100		
11					

Below is the final function: The task can be accomplished with the following **OR()** function:

```
=OR(B2="Water",C2>60)
```

	A	B	C	D	E
1	Name	Type 1	Defense	Water Type OR more than 60 Defense	
2	Bulbasaur	Grass	49	=OR(B2="Water"; C2>60	
3	Ivysaur	Grass	63	OR (logical1; [logical2]; [logical3]; ...)	
4	Venusaur	Grass	83		
5	Charmander	Fire	43		
6	Charmeleon	Fire	58		
7	Charizard	Fire	78		
8	Squirtle	Water	65		
9	Wartortle	Water	80		
10	Blastoise	Water	100		

Since the value in cell B2 is not **"Water,"** the first condition is **FALSE**.

Since the value in cell C2 is **less than 60, the second** condition is also **FALSE**.

At least one condition needs to be **TRUE** for the OR() function to return **TRUE**.

The function can be repeated with the **filling function** for each row to perform the same check for each Pokémon:

	A	B	C	D	E
1	Name	Type 1	Defense	Water Type OR more than 60 Defense	
2	Bulbasaur	Grass	49	=OR(B2="Water"; C2>60)	
3	Ivysaur	Grass	63	=OR(B3="Water"; C3>60)	
4	Venusaur	Grass	83	=OR(B4="Water"; C4>60)	
5	Charmander	Fire	43	=OR(B5="Water"; C5>60)	
6	Charmeleon	Fire	58	=OR(B6="Water"; C6>60)	
7	Charizard	Fire	78	=OR(B7="Water"; C7>60)	
8	Squirtle	Water	65	=OR(B8="Water"; C8>60)	
9	Wartortle	Water	80	=OR(B9="Water"; C9>60)	
10	Blastoise	Water	100	=OR(B10="Water"; C10>60)	
11					

Now, each row is checked for `**Water**` type or **Defense greater than 60**:

	A	B	C	D	E
1	Name	Type 1	Defense	Water Type OR more than 60 Defense	
2	Bulbasaur	Grass	49	FALSE	
3	Ivysaur	Grass	63	TRUE	
4	Venusaur	Grass	83	TRUE	
5	Charmander	Fire	43	FALSE	
6	Charmeleon	Fire	58	FALSE	
7	Charizard	Fire	78	TRUE	
8	Squirtle	Water	65	TRUE	
9	Wartortle	Water	80	TRUE	
10	Blastoise	Water	100	TRUE	
11					

For example:

- **Ivysaur** is not a **`Water`** type, but it has more **than 60 defenses**, so the function returns **"TRUE."**
- **Charmeleon** is not a **`Water`** type and has less **than 60 defense**, so the function returns **"FALSE."**
- **Wartortle** is both a **`Water`** type and has more **than 60 defenses**, so the function returns **"TRUE."**

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	A	B	C	D
1	Name	Type 1	Speed	Water Type OR more than 60 Speed
2	Bulbasaur	Grass	45	FALSE
3	Ivysaur	Grass	60	FALSE
4	Venusaur	Grass	80	TRUE
5	Charmander	Fire	65	TRUE
6	Charmeleon	Fire	80	TRUE
7	Charizard	Fire	100	TRUE
8	Squirtle	Water	43	TRUE
9	Wartortle	Water	58	TRUE
10	Blastoise	Water	78	TRUE
11				
12	Step 1. `=OR(B2="Water",C2>60)			
13	*similar to AND			
14	*Returns TRUE if ONE or both conditions are met			
15	*When none of the conditions are met returns FALSE			

4.2: Example OR() Function with IF()

In Excel, the **OR()** function itself returns either **TRUE** or **FALSE** and these return values CANNOT be changed. However, when you use **AND()** within an **IF()** function, you can customize the output based on the result of the AND() function. For example, you can return "YES" or "NO," "Open" or "Close," or any other values you choose.

In this example we will use the **4.2_OR** sheet from our sample file.

Below is the final function: The task can be accomplished with the following **OR()** function.

	A	B	C	D	E	F
1	Name	Type 1	Defense	Water Type OR more than 60 Defense		
2	Bulbasaur	Grass	49	=IF(OR(B2="Water"; C2>60); "Yes"; "No")		
3	Ivysaur	Grass	63			
4	Venusaur	Grass	83			
5	Charmander	Fire	43			
6	Charmeleon	Fire	58			
7	Charizard	Fire	78			
8	Squirtle	Water	65			
9	Wartortle	Water	80			
10	Blastoise	Water	100			
11						

The function can be repeated with the **filling function** for each row to perform the same check for each Pokémon, as shown in the screenshot below.

	A	B	C	D	E
1	Name	Type 1	Defense	Water Type OR more than 60 Defense	
2	Bulbasaur	Grass	49	=IF(OR(B2="Water"; C2>60); "Yes"; "No")	
3	Ivysaur	Grass	63	=IF(OR(B3="Water"; C3>60); "Yes"; "No")	
4	Venusaur	Grass	83	=IF(OR(B4="Water"; C4>60); "Yes"; "No")	
5	Charmander	Fire	43	=IF(OR(B5="Water"; C5>60); "Yes"; "No")	
6	Charmeleon	Fire	58	=IF(OR(B6="Water"; C6>60); "Yes"; "No")	
7	Charizard	Fire	78	=IF(OR(B7="Water"; C7>60); "Yes"; "No")	
8	Squirtle	Water	65	=IF(OR(B8="Water"; C8>60); "Yes"; "No")	
9	Wartortle	Water	80	=IF(OR(B9="Water"; C9>60); "Yes"; "No")	
10	Blastoise	Water	100	=IF(OR(B10="Water"; C10>60); "Yes"; "No")	
11					

Now, each row has a check for `**Water**` Type or **Defense greater than 60**.

	A	B	C	D	E
1	Name	Type 1	Defense	Water Type OR more than 60 Defense	
2	Bulbasaur	Grass	49	No	
3	Ivysaur	Grass	63	Yes	
4	Venusaur	Grass	83	Yes	
5	Charmander	Fire	43	No	
6	Charmeleon	Fire	58	No	
7	Charizard	Fire	78	Yes	
8	Squirtle	Water	65	Yes	
9	Wartortle	Water	80	Yes	
10	Blastoise	Water	100	Yes	
11					

For example:

- Ivysaur is not a Water type, but it has more than 60 defenses, so the function returns "Yes."

- Charmeleon is not a **"Water"** type and has less than 60 defense, so the function returns "No."
- Wartortle is both a **"Water"** type and has more than 60 defense, so the function returns "Yes."

MY SCREENSHOT

	A	B	C	D
1	**Name**	**Type 1**	**Speed**	**Water Type OR more than 60 Speed**
2	Bulbasaur	Grass	45	**FALSE**
3	Ivysaur	Grass	60	**FALSE**
4	Venusaur	Grass	80	**FALSE**
5	Charmander	Fire	65	**FALSE**
6	Charmeleon	Fire	80	**FALSE**
7	Charizard	Fire	100	**FALSE**
8	Squirtle	Water	43	**FALSE**
9	Wartortle	Water	58	**FALSE**
10	Blastoise	Water	78	**FALSE**
11				
12	Step 1. `=IF(OR(B2="Water",C2>60),"Yes"<"No")			
13	*OR is nested into the IF function where the logical function normally goes			
14	*Used to customize output			

Reference:

https://support.microsoft.com/en-us/office/Formulas-and-functions-294d9486-b332-48ed-b489-abe7d0f9eda9

Submission Guidelines:

- Include the following deliverables in your submission:

- Take a screenshot of your worksheets, Submit your source file, including screenshot, using the **Start Assignment** button in the top-right corner of the Assignment page in Canvas.
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